

CSE 316

ASSIGNMENT 3

**Report on finding max cut of a graph using GRASP and
explanation of output**

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Objective:

To construct a max cut solution (two sets of vertices) from a graph and improve the solution using GRASP (greedy randomized adaptive search procedure).

Working Procedure:

GRASP uses multistart local search procedure where the local search starts from random position to overcome the problems of local search. This way solution can be improved using GRASP. Each GRASP iteration has construction phase and local search phase.

Construction:

During construction phase, we generate an initial solution consist of two sets of vertices by partitioning the graph's vertices using four different approaches such as greedy, semi greedy and two types of randomized approaches.

Greedy:

Here, we begin by selecting an edge with maximum weight. Then we place the two endpoints of this edge into the different sets. After that, from the remaining vertices, we choose a vertex and assign it to one of the sets. Then we calculate the maximum cut value considering the two sets. Then we repeat the same process for the same vertex but by placing the vertex in the other set. Then we store the max cut values for the vertex. This procedure is repeated for all the remaining vertices and the vertex with the max value is selected and placed into the appropriate set that corresponds to this maximum value. This way we include all the vertices in the two distinct sets.

Semi greedy:

Initially, we generate a restricted candidate list (RCL) through the utilization of value based method. To generate the list, a threshold, calculated as $(\min + \alpha(\max - \min))$ is used and all the edges greater than this threshold is placed in the list. Here, the value of ' α ' is generated randomly. Then we randomly select the first edge from RCL and place two end points of this edge in two different sets.

Next, from all remaining vertices, we choose a vertex and place in one of the sets. Then we calculate max cut between the two sets. Then for this vertex we repeat the same procedure by placing the vertex in the other set. We store the max cut values in two separate maps. We do this procedure for all the remaining vertices. After that, we calculate max value and min value from the maps. We use

this values to generate new threshold. Then from remaining vertices which have values greater than or equal to calculated threshold, we generate a candidate list. From the candidate list we randomly select a vertex and place it into the appropriate set for which cut value is maximized. This way we include all the vertices in the sets.

Randomized:

In the 'Random' approach, we select vertices at random and assign it in one of the sets with a probability of 0.5.

In the 'Random2' approach, we mainly use the semi greedy approach with alpha equal to '0'. As a result, the resulting solution consists of a random selection of vertices for both sets.

After the construction phase, we use local search to improve the constructed solution.

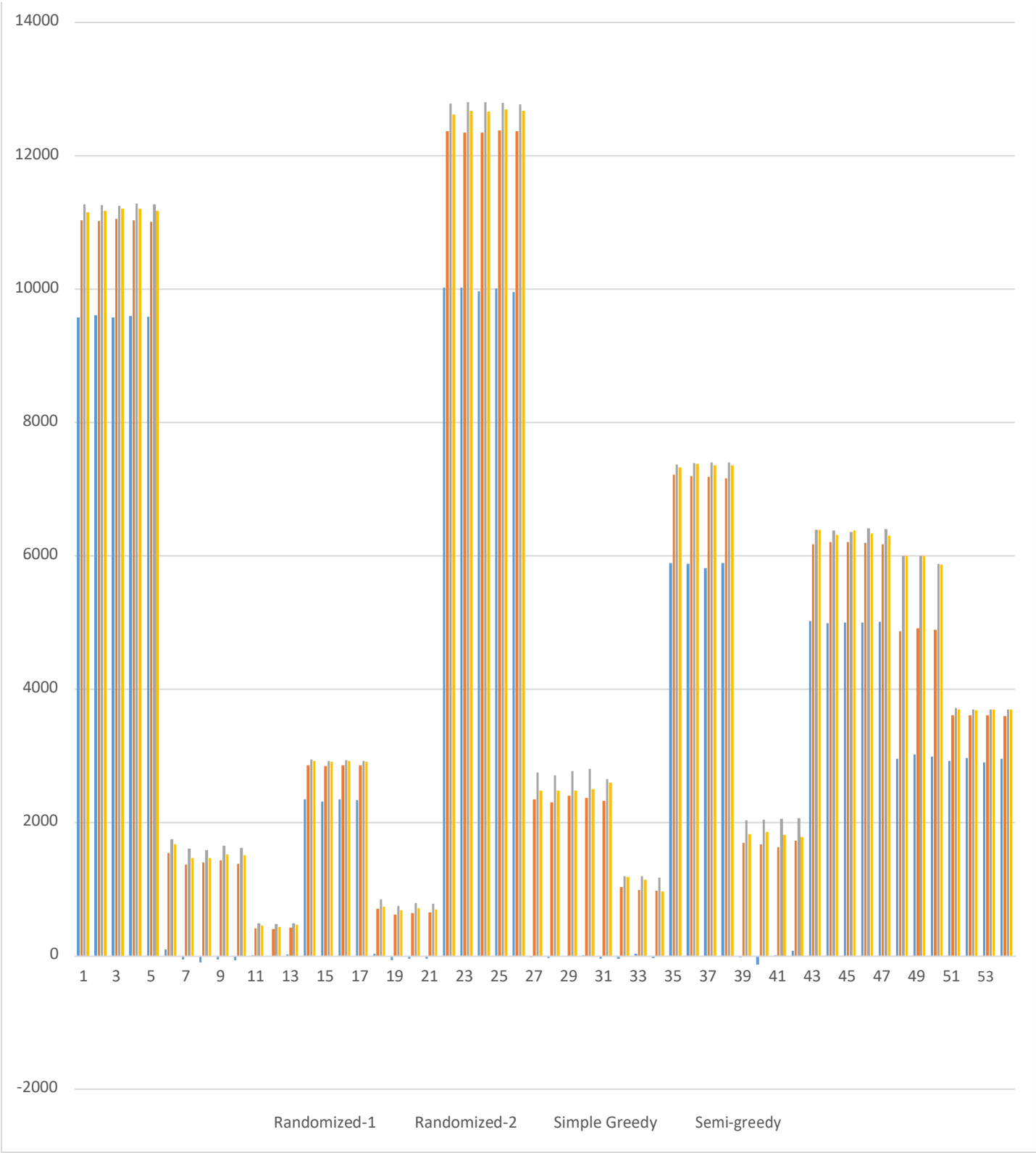
Local Search:

From the solution sets, we choose a vertex and place the vertex in other set. Then we calculate max cut. If result is improved, then this will be our new solution. We apply this procedure till we reach a local maxima.

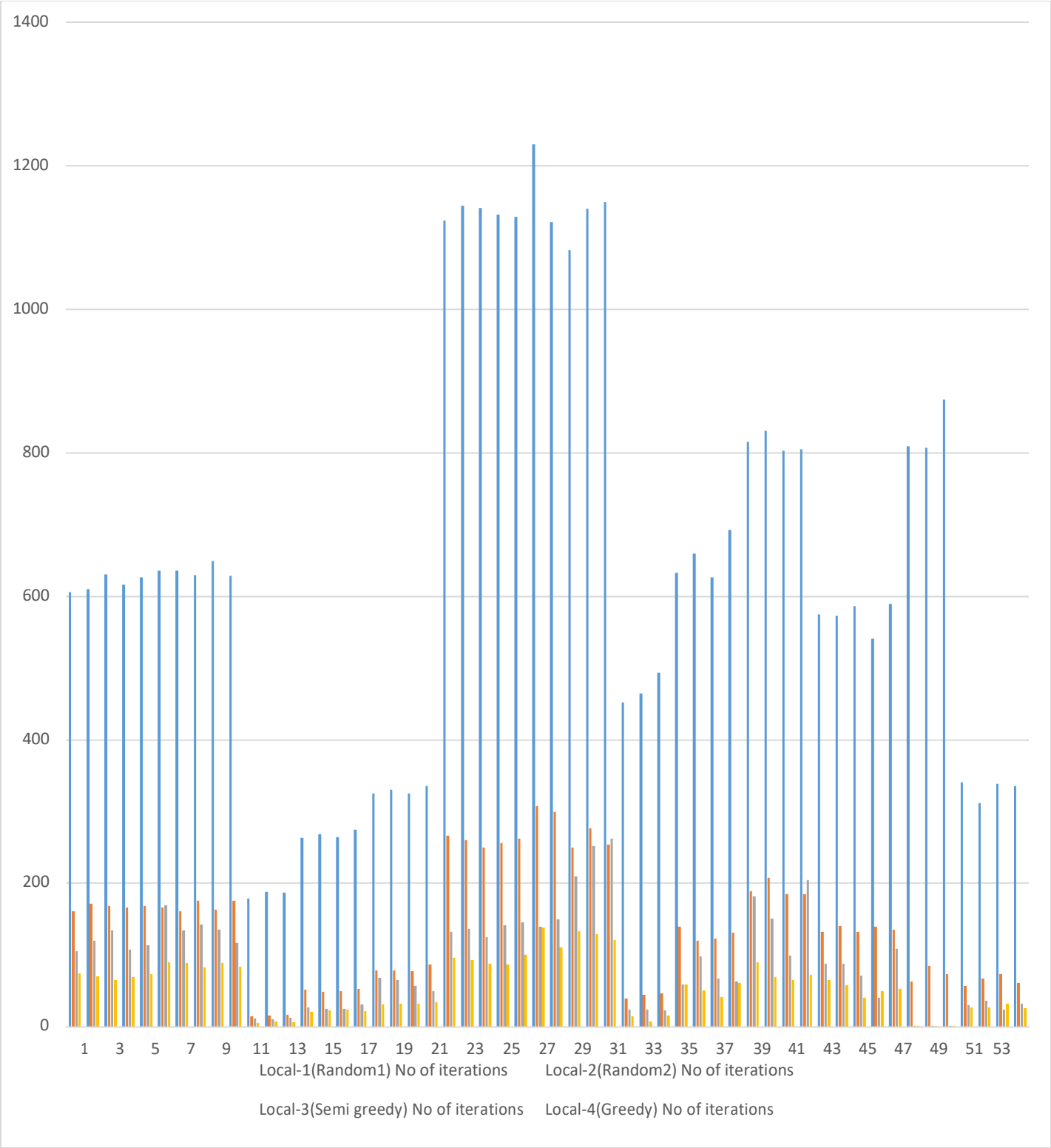
In GRASP, we execute construction phase and local search phase for a specific number of iterations and we select the best result from all the iterations.

(All of the algorithms are either from 'Optimization by GRASP' written by Mauricio G.C. Resende & Celso C. Ribeiro or constructed from assignment specifications)

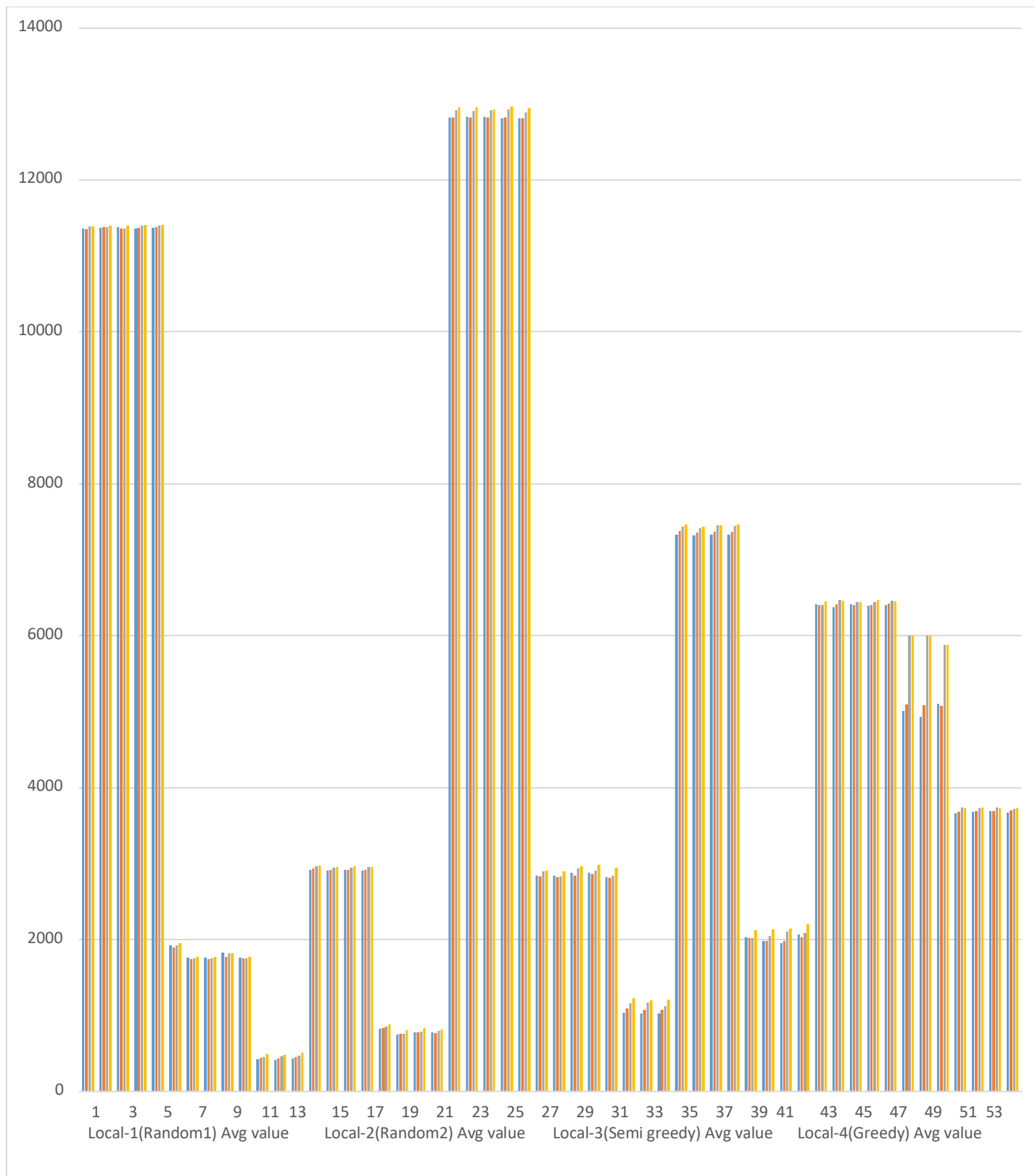
Graphs:



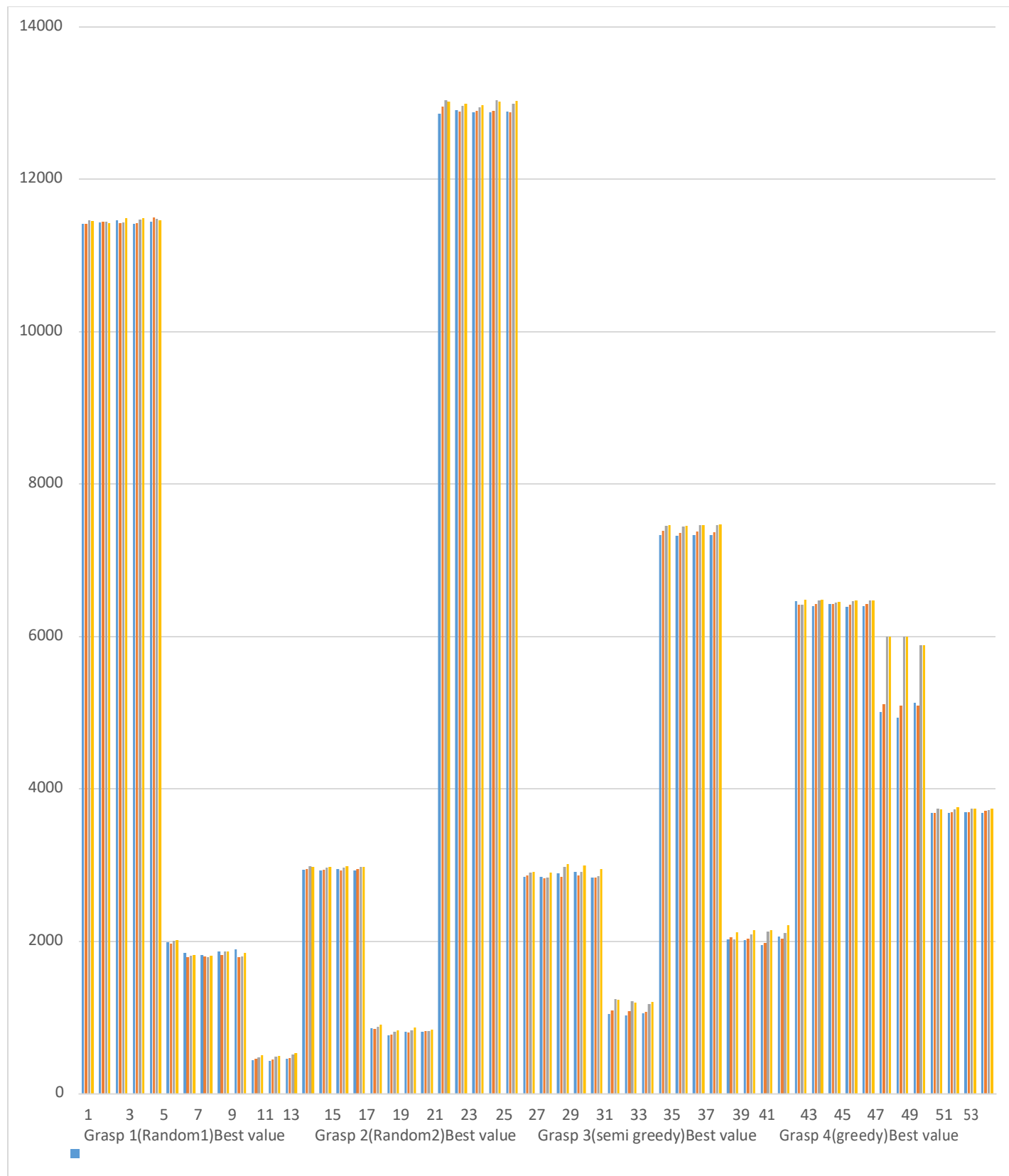
Graph.1: Comparison of max cut values between different construction approaches



Graph.2: Comparison of iterations between different local search approaches



Graph.3: Comparison of max cut values between different local search approaches



Graph.4: Comparison of max cut values between different GRASP approaches

Analysis of Output:

After running 54 test cases we have found:

- The randomized constructions are not so good. In fact, it often returns negative max cut value for the graphs which consist of negative edges.
- The randomized approach 1 performs poorly. But approach 2 is not so bad. Approach 2 gives max cut value that is greater than or equal to 60% of the upper bound.
- The greedy construction algorithm gives max cut value which is closer to upper bound. Most of the time, the value is greater than or equal to 85% of the upper bound.
- The semi greedy construction algorithm sometime performs well sometime not. But it is better than randomized approaches.
- Comparatively, greedy construction algorithm performs better than semi greedy construction algorithm in about 80% cases.
- In almost 99% cases, greedy and semi greedy performs better than randomized construction algorithms.
- After doing local search over randomized approach 1 and approach 2, both of searches gives result that is almost same. But for approach 1 local search takes more iterations than approach 2.
- Local search over randomized constructions take most iterations whereas that of greedy construction takes least iterations.
- Local search over greedy and semi greedy constructions give better max cut value than that of randomized constructions.
- After applying GRASP, max cut value of all of the constructions are improved. But for semi greedy construction, in almost 70% cases the improvement is higher than any other constructions.
- Overall, final result obtained from GRASP is greater than or equal to 90% of the upper bound.